

TREPP: Tandem Repeat Expansion Pathogenicity Predicting Approach Using Stacked CatBoost Models and Multiple Features

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Abstract. Excessive expansion of tandem repeats (TRs) increases the risk of various diseases, including cancer, autism, and neurodegenerative disorders. Accurately predicting the pathogenicity of TR expansions is challenging due to their dependence on repeat length, motif repeat threshold, and the dynamic nature of repeat changes. The existing method primarily uses association information between TR loci and genes, neglecting sequence characteristics that contribute to TR instability. To address these challenges, we introduce TREPP, a machine learning-based framework designed to predict high-risk pathogenic TR loci. To improve the predictive performance, TREPP integrates multi-dimensional features, including flanking sequence composition, GC content, and the density of surrounding TR regions. At the same time, the lack of validated pathogenic TRs results in a severe imbalance between positive and negative samples. To solve this problem, we employ a multi-sampling strategy and train multiple CatBoost models through ensemble learning to mitigate data imbalance. Test on a manually curated pathogenic TR dataset, TREPP achieves an AUPRC of 91.57%, increases 4.17% over the state-of-the-art method RExPRT. Additionally, in the OMIM and GeneTrek disease gene databases, TREPP identifies more TR-associated pathogenic genes than RExPRT, demonstrating its comprehensive predictive capability.

Keywords: Tandem Repeat · Pathogenicity Prediction · CatBoost · Ensemble Learning

1 Introduction

The human genome contains over one million discrete tandem repeats (TRs), accounting for approximately 3% of its total sequence [1, 2]. These TRs include short tandem repeats (STRs) with motifs of 1–6 base pairs and variable number

tandem repeats (VNTRs) with motifs exceeding 7 base pairs [3, 4]. Notably, TRs exhibit mutation rates higher than other genetic variations, such as single nucleotide polymorphisms (SNPs) [5]. This hypermutability stems from a critical length threshold phenomenon: when the number of motifs exceeds 3-5 repeat units, DNA replication slippage rates increase exponentially [6]. Such unstable expansions can lead to rapid TR length changes, potentially causing genomic instability and subsequent pathological consequences [2].

To date, STR expansions have been identified as the cause of 65 nervous system diseases and 14 neuromuscular diseases, collectively known as repeat expansion disorders [7, 8]. Repeat expansion disorders exhibit diverse pathogenic mechanisms, which depend on the location of the amplified TR within a gene locus as well as the function of the affected gene, including loss of function, gain of function, or dominant-negative effects [9]. These mechanisms are influenced by STR length, genomic context, sequence motifs, and the extent of motif expansion [10]. Since multiple mechanisms can act simultaneously and contribute to disease progression, identifying the precise pathogenic mechanism remains challenging. Beyond their pathogenic effects, expanded loci often exhibit sequence instability, with repeat sizes progressively increasing across generations [10]. Accurately predicting the potential pathogenicity of TR expansions is crucial for early diagnosis and disease intervention.

Despite their functional significance, TR expansions have historically been underrepresented in genetic studies due to persistent technical and analytical challenges. Recent studies have underscored their significant role in human diseases, leading to an increasing number of computational approaches for pathogenicity prediction [10–12]. A major challenge in detecting TR expansion pathogenicity is the significant class imbalance in available datasets. Since the pathogenic loci requires experimental validation, the STRchive [13] database currently contains only 68 verified TR loci associated with diseases. Most standard machine learning algorithms struggle with such datasets, as they tend to favor the majority class, leading to poor predictive accuracy for the minority class [14]. Boosting, one of the most widely used ensemble learning techniques, constructs a strong classifier by combining many weak classifiers [15]. Notably, the popular boosting methods, including XGBoost [16], LightGBM [17], and CatBoost [18], have been extensively applied in the prediction and prognosis of genetic disorders [19, 20].

RExPRT [21] is the state-of-the-art method for predicting the pathogenicity of TR expansions based on machine learning. It classifies TRs as benign or pathogenic by analyzing their genomic annotation features, such as constraint scores, overlapping regulatory regions, and transcription factor binding sites. However, it still falls short in sensitivity due to the neglect of key genomic context. Here, we present TREPP, a novel algorithm for predicting TR expansion pathogenicity. TREPP uses the flanking regions of TRs and the positional relationship between genes and TR loci to enhance accuracy and generalization. To address class imbalance, TREPP employs a multi-model ensemble approach, training multiple CatBoost base models on diverse negative sample subsets and

combining their predictions through a logistic regression meta-learner. Benchmarking results show that TREPP significantly outperforms RExPRT, demonstrating its superior accuracy in pathogenic TR prediction. TREPP is publicly accessible on GitHub (<https://github.com/minghuaxu/TREPP>).

2 Methods

As shown in Fig. 1, TREPP uses both the gene and sequence features for each candidate TR locus to improve the performance. These features are combined and fed into multiple CatBoost-based models for initial classification. The initial predictions from these base models are then integrated using a stacking approach, where they serve as input features for a logistic regression model. The logistic regression model ultimately calculates the pathogenicity probability of the TR locus, generating binary predictions (0 for non-pathogenic, 1 for pathogenic).

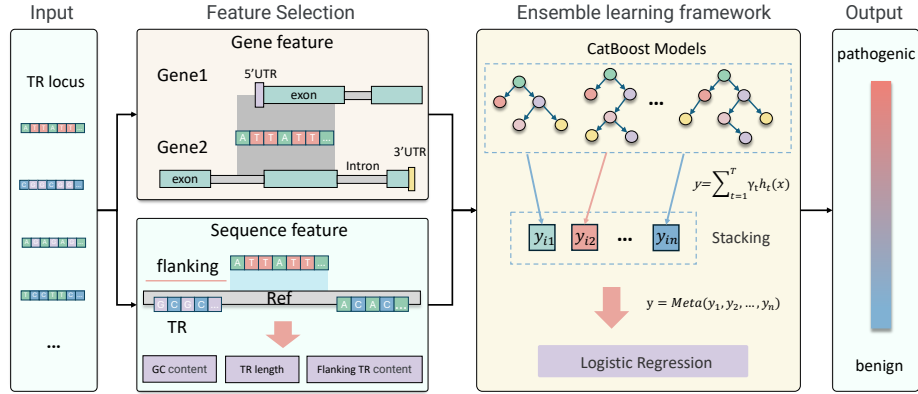


Fig. 1. The workflow of TREPP. Gene and flanking sequence features of TR loci are encoded into feature vectors, followed by the prediction of TR pathogenicity using the ensemble learning method.

2.1 Feature Extraction

Our feature extraction framework captures two fundamental characteristics for pathogenic TR prediction: TR-gene positional relationships and local genomic context features.

TR-Gene Positional Relationships We quantify spatial associations between TR loci and functional genomic elements using GENCODE annotations [22]. For each TR locus $TR_i = (c, s_i, e_i)$ (chromosome c , positions s_i to e_i) and genomic feature $G_j = (c, s_j, e_j)$, we compute a binary overlap indicator:

$$F_{ij} = \begin{cases} 1, & \text{if } c = c_{G_j} \text{ and } [s_i, e_i] \cap [s_j, e_j] \neq \emptyset \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where overlap requires both chromosomal identity and coordinate intersection.

We analyze four categories of genomic features: (1) gene bodies with ± 5 kb flanking regions, (2) transcripts with ± 5 kb flanking regions, (3) coding regions including 5'UTR and 3'UTR with ± 500 bp flanking sequences, and (4) regulatory elements (promoters and terminators) with ± 500 bp flanking sequences.

Local Genomic Context Features We characterize sequence properties through three complementary approaches. First, GC content is calculated for three distinct regions: the TR core region itself, its immediate 500 bp flanking regions, and broader 5000 bp flanking regions. Second, sequence complexity is quantified using tandem repeat density:

$$\text{TR Density}(F) = \frac{\sum_{r \in R(F)} \text{len}(r)}{\text{len}(F)} \quad (2)$$

where $R(F)$ represents all tandem repeats detected in flanking region F using Tandem Repeats Finder (TRF) [23] with the parameter set "2 7 80 10 30 2000 -L 2 -H".

Third, we extract TR intrinsic features including three key aspects: the length of the TR locus, the GC composition of the TR sequence, and the nucleotide composition of the repeat motif. This integrated feature set provides comprehensive characterization of both the genomic context and sequence properties of TR loci at multiple scales, from the immediate repeat sequence to distal flanking regions.

2.2 Model Training

After feature extraction, we employ CatBoost with an ensemble strategy to address data imbalance and improve model stability for robust pathogenicity prediction. CatBoost, a gradient boosting decision tree (GBDT) algorithm, handles categorical features natively without requiring additional preprocessing. Its fundamental formula can be expressed as:

$$f(x) = \sum_{t=1}^T \gamma_t h_t(x) \quad (3)$$

where $h_t(x)$ represents the prediction of the t -th tree, γ_t denotes the corresponding learning rate or weight, and T is the total number of trees.

The model mitigates data imbalance and small-sample challenges by utilizing a symmetric tree structure and a user-defined loss function to capture characteristics of the minority class. Its efficient gradient boosting mechanism ensures

robust performance on small datasets. To further enhance stability and generalization, we divide the negative samples in the training set into 15 equal subsets. Each subset is combined with all positive samples to create 15 balanced training subsets. We then train 15 independent CatBoost base models, ensuring that each model retains the original feature distribution while reducing bias caused by class imbalance. This ensemble strategy maximizes model diversity, thereby minimizing the risk of bias and overfitting.

Then, a stacking-based ensemble learning strategy is adopted, where predictions from the 15 CatBoost models serve as meta-features and are fed into a logistic regression model for final integration. We combine the outputs of n CatBoost models. Specifically, let y_i denote the prediction from the i -th model for $i = 1, 2, \dots, n$. The final prediction is obtained by applying a meta-model that aggregates these outputs as follows:

$$y = \text{Meta}(y_1, y_2, \dots, y_n) \quad (4)$$

This hierarchical framework captures comprehensive feature patterns, enhancing the model’s representational power by aggregating diverse predictions from multiple base models. The integration strategy improves prediction robustness and stability by optimally combining model outputs through the meta-learner, thereby boosting overall performance. The final output is a reliable pathogenicity score, providing a more accurate assessment of TR pathogenicity.

Finally, we employ Optuna [24] for automated hyperparameter tuning, optimizing CatBoost and logistic regression parameters to enhance predictive performance. Optuna is an efficient, automated optimization tool that utilizes Bayesian optimization and the tree-structured Parzen estimator (TPE) to systematically search for the best hyperparameter configurations. For each CatBoost model, key hyperparameters including tree depth, learning rate, and regularization coefficients are fine-tuned to optimize model performance. In the ensemble stage, we also optimize the logistic regression model’s regularization strength (L1 or L2). By leveraging the automated search strategy of Optuna, we efficiently identify optimal hyperparameter settings, significantly improving predictive performance.

3 Results

3.1 Performance Evaluation

The dataset used in this study is derived from the manually curated data of the RExPRT. TR loci with fully validated causal relationships to diseases are labeled as pathogenic, and those are commonly expanded in the 1000 Genomes Project controls as negative training data[25]. Since control TR loci usually show expansion of more than 150 bp in healthy genomes, those loci are considered as benign. The dataset consists of 60 pathogenic TR loci and 700 benign control samples, resulting in a significant class imbalance. To ensure a fair comparison of the performance of different tools, we follow the same training and test set

partitioning method as RExPRT. Specifically, the test set contains the same 20 pathogenic TR samples and 83 benign control samples as RExPRT, while the remaining samples are used for training.

Compared to the state-of-the-art method RExPRT, TREPP demonstrates significant improvements across multiple evaluation metrics, particularly in scenarios with severe class imbalance. As shown in Fig. 2a, TREPP achieves a precision of 0.9770, a recall of 0.9048, and an F1-score of 0.9356, representing increases of 3.51%, 2.99%, and 3.22% over RExPRT, respectively. Given the substantial imbalance between positive and negative samples, we evaluate model performance using AUROC and AUPRC. As shown in Fig. 2b and c, TREPP achieves an AUROC of 0.9558 and an AUPRC of 0.9277, exceeding RExPRT by 2.29% and 4.17%, respectively.

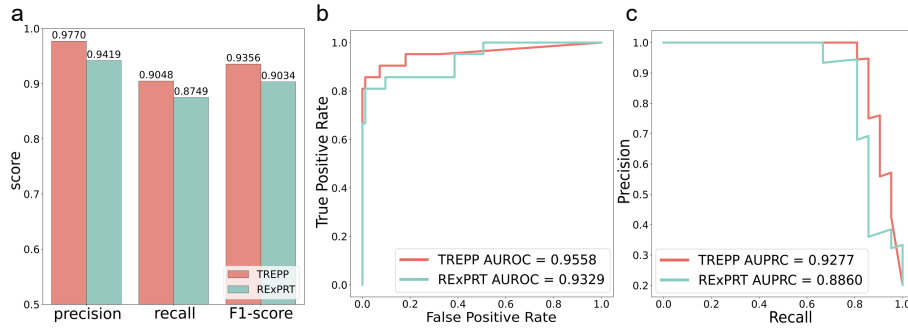


Fig. 2. Performance evaluation of TREPP and RExPRT.

At the same time, TREPP mitigates bias introduced by class imbalance through its ensemble learning strategy. By training multiple base models on balanced subsets, TREPP captures minority class characteristics more effectively than RExPRT, leading to higher recall without sacrificing precision. Notably, TREPP exhibits improved detection of pathogenic TR loci within intronic regions, where existing methods often struggle due to complex sequence structures and regulatory variations. Through integration of gene-level and sequence-level features, TREPP demonstrates robust predictive performance in challenging genomic contexts.

3.2 Explaining Model with SHAP

To interpret our model’s decision-making process, we implement SHAP (SHapley Additive exPlanations) to analyze feature importance in each CatBoost base model. SHAP values quantify the impact of specific genomic attributes on predictions, providing mechanistic insights into disease risk determinants. Fig. 3a displays the global feature importance plot, which ranks features by their aggregated contribution magnitude. The five most influential features include:

TR_Content_1k (proportion of other TR content within the 1k flanking sequence), Exon (whether the TR locus is located in an exon), TR_Len (length of the TR site itself), GC_Content_1k (GC content within the 1k flanking sequence), and Motif_GC (GC content composition of the TR site).

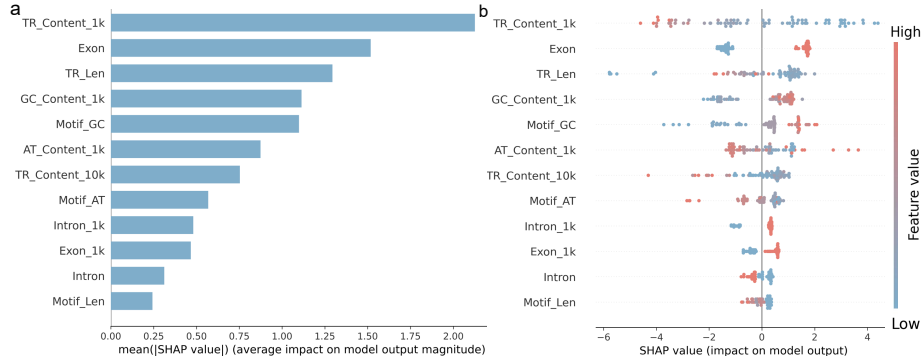


Fig. 3. Explanation of CatBoost models using SHAP. a. The SHAP summary plot for a single CatBoost base model demonstrates the contribution of each feature to prediction outcomes. b. The SHAP dependence plot for a single CatBoost base model reveals the relationship between feature values and their corresponding SHAP values.

To further explore the relationship between key features and TR pathogenicity, we analyze SHAP dependency plots for a single CatBoost-based model (Fig. 3b). These plots illustrate feature interactions, with the color of each point representing interacting variables, providing insights into how specific features drive model predictions. The SHAP values indicate that TR_Content_1k, TR_Len, and GC_Content_1k are the main negative factors, while Exon, GC_Content_1k, and Motif_GC are the main positive factors. When TRs are located in exons, their amplification behavior is more likely to disrupt gene function. Additionally, high GC content is prone to slippage during DNA replication, making the replicated sequence more likely to amplify, which is consistent with existing research [26–28]. Interestingly, our analysis reveals an inverse relationship between TR pathogenicity and the density of surrounding TRs. Specifically, TR loci surrounded by fewer TRs exhibit a higher risk of pathogenicity. We speculate that multiple repeat regions may form secondary structures (e.g., hairpin loops or cross-structures) during DNA replication [29], causing replication forks to stall or break. If fewer repeats are present in the surrounding region, only the repeats at the TR site are likely to form a structure, concentrating replication pressure at this site and making it more susceptible to amplification.

3.3 Ablation Experiment

To evaluate the contribution of each feature, we implement three models: All (full feature set), All-gene (excluding gene-related features), and All-seq (excluding

sequence-based features). As shown in Fig. 4a, excluding any feature leads to a decline in precision, recall, and F1-score. The removal of the gene feature results in a decrease of model in the precision from 0.9770 to 0.8482 (-13.2%), recall from 0.9048 to 0.8092 (-10.6%) and F1-score from 0.9356 to 0.8264 (-11.7%).

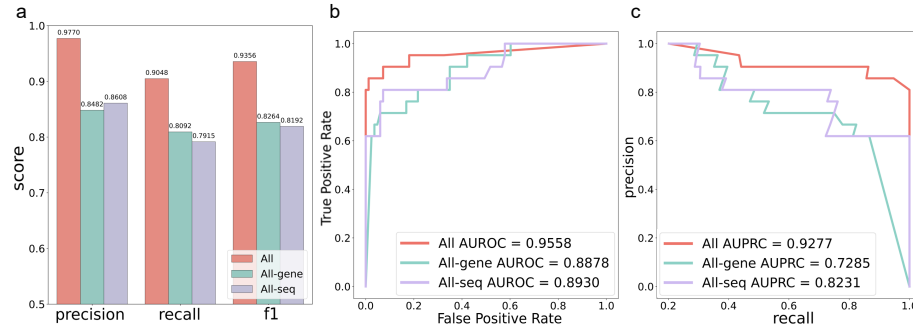


Fig. 4. The predictive performance of TREPP with different features.

The removal of the sequence feature results in a decrease of model in the precision from 0.9770 to 0.8608 (-11.9%), recall from 0.9048 to 0.7915 (-12.5%) and F1-score from 0.9356 to 0.8192 (-12.4%). The 11.64% absolute reduction in F1-score following the removal of sequence features demonstrates their critical contribution to model performance. As shown in Fig. 4b and c, the model achieves peak discriminative performance (AUROC: 0.9558 and AUPRC: 0.9277) only when retaining all features. The results confirm that all features contribute significantly to the prediction performance.

3.4 Gene Overlap Analysis

We compare the overlap between genes linked to pathogenic TR loci detected by the RExPRT and TREPP in the human GRCh37 genome using OMIM[30] (Online Mendelian Inheritance in Man) and GeneTrek[31]. OMIM serves as a comprehensive knowledgebase for Mendelian disorders, encompassing over 16,000 genes with manually curated annotations spanning phenotypic profiles, inheritance patterns, pathogenic variants, and molecular mechanisms. These data are systematically curated from peer-reviewed studies, clinical case reports, and expert consensus. GeneTrek specializes in neurodevelopmental disorder(NDD)-associated genes, aggregating evidence from large-scale whole-exome sequencing (WES) and whole-genome sequencing (WGS) cohort studies.

GENCODE annotations are used to obtain associations between TR loci and genes, defining the link as overlap between a TR locus and a gene genomic coordinates. As shown in Fig. 5, both TREPP and RExPRT detect a significant number of pathogenic genes that overlap with the public databases OMIM and GeneTrek. However, TREPP identifies more pathogenic genes than

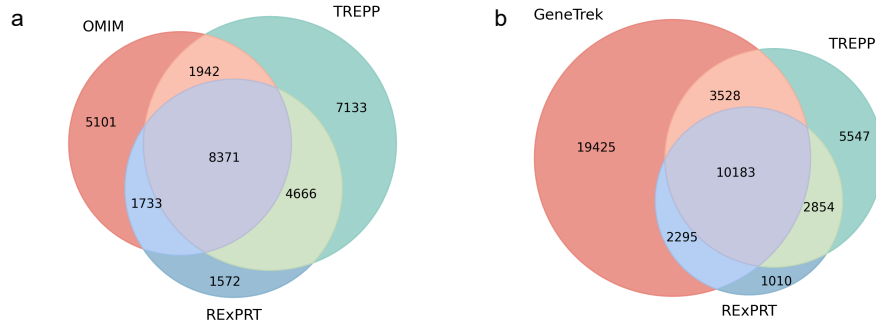


Fig. 5. Overlap between RExPRT and TREPP, and pathogenic gene databases: a. OMIM. b. GeneTrek.

RExPRT. For example, TREPP, RExPRT, and OMIM share 8,371 pathogenic genes, while the unique overlap between TREPP and OMIM is 1,942, surpassing the unique overlap of 1,733 between RExPRT and OMIM (Fig. 5a). Similarly, TREPP, RExPRT, and GeneTrek share 10,183 pathogenic genes, with TREPP and GeneTrek showing a unique overlap of 3,528, compared to 2,295 for RExPRT and GeneTrek (Fig. 5b). These results demonstrate that TREPP achieves higher sensitivity in detecting pathogenic genes compared to RExPRT. It is worth noting that while the public databases OMIM and GeneTrek provide curated results, their coverage of disease-related genes is still limited. Automated prediction tools such as TREPP and RExPRT can identify pathogenic genes that are not widely recognized, potentially shedding light on the mechanisms of understudied diseases.

3.5 Genome-Wide Distribution of TR Loci

TREPP is also applied to analyze the genome-wide distribution of high-risk pathogenic TR loci. From 829,222 candidate TRs with motif lengths ranging from 3 to 8 bp in the hg19 reference genome, TREPP predicts 39,882 pathogenic TR loci. The results reveal that these high-risk pathogenic TRs are primarily enriched in functional genomic regions. As shown in Fig. 6a, pathogenic TRs are predominantly enriched in regulatory elements and their adjacent genomic regions. The expansion of these TRs may disrupt cis-regulatory elements, potentially leading to gene expression dysregulation and disease. In contrast, non-pathogenic TRs are commonly located within intronic regions, where their expansion typically has minimal impact on gene function and structure.

To further investigate chromosomal distribution, chromosomes are divided into 1 Mb bins to visualize the density of pathogenic TR loci (Fig. 6b). Although the distribution of pathogenic loci is relatively uniform across chromosomes, we observe a significant enrichment in the telomeric regions of certain chromosomes, with color intensity positively correlating with locus density (red indicating higher density). Existing literature suggests that GC-rich sequences

are more prone to slippage mismatches during DNA replication, leading to an increased frequency of amplification events [26–28]. This observation is supported by our findings, as pathogenic TR motifs exhibit significantly higher GC content (Fig. 6c and d).

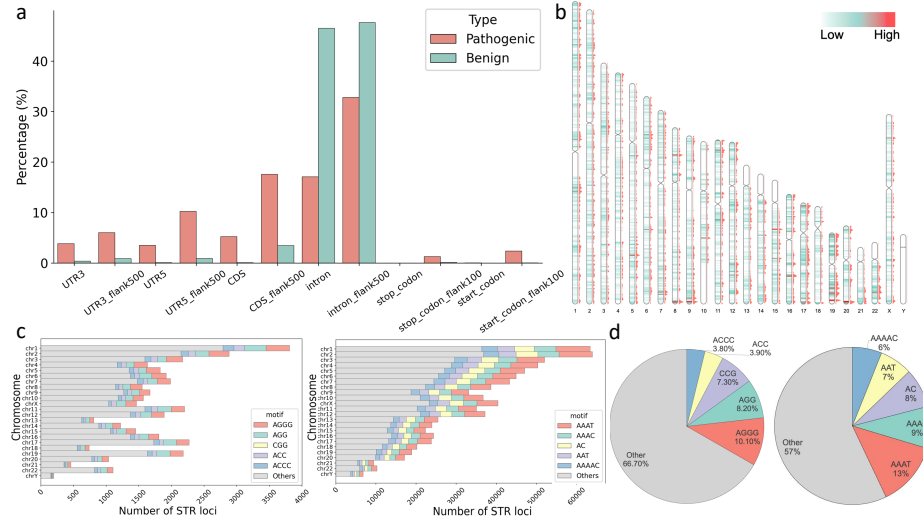


Fig. 6. The genome-wide distribution characteristics of high-risk pathogenic TRs. a. The distribution pattern of TR loci within genomic regulatory elements. b. The chromosomal enrichment of pathogenic TRs. c. The differences in motif composition between pathogenic (left) and non-pathogenic (right) TRs across chromosomes. d. The five most common motif base compositions in pathogenic (left) and non-pathogenic (right) TRs.

4 Conclusion

Tandem repeats have been recognized as critical contributors to neurodegenerative disorders. While existing tools like RExPRT demonstrate progress in TR pathogenicity prediction through genomic annotation analysis and machine learning applications, current methodologies face three primary limitations: (1) DNA repeat region sequence complexity is not considered, (2) the lack of validated pathological cases leads to extremely unbalanced positive and negative samples, and (3) limited model interpretability. In this study, we propose TREPP, a novel tool designed to predict TR expansion pathogenicity through a multi-dimensional feature integration and ensemble learning approach. TREPP incorporates the positional relationships between TR loci and neighboring genes, the complexity of the flanking region sequence, and variations in the content of GC. Moreover, TREPP implements a multi-sampling strategy, generating multiple balanced training subsets by pairing all positive samples with different subsets

of negative samples. We optimize and train multiple CatBoost models using Optuna, whose predictions are then stacked and used as input for a logistic regression model that generates the final pathogenicity scores.

TREPP achieves higher precision, recall, and F1-score than RExPRT in the test data set. Ablation experiments confirm that gene annotation and sequence features are critical for predicting TR pathogenicity. SHAP value analysis also identifies key factors driving TR pathogenicity, including TR content in flanking regions, TR GC content, and its overlap with exons. TREPP also identifies a higher number of pathogenic TR-associated genes throughout the genome compared to RExPRT, as verified by overlap with the OMIM and GeneTrek databases, underscoring its sensitivity in detecting disease-relevant TRs. Furthermore, we applied TREPP to the GRCh37 genome to identify pathogenic TRs and map their distribution, providing valuable information on potential disease associations. In summary, TREPP provides a more robust and interpretable approach to TR pathogenicity prediction, offering new insights for studying TR-associated diseases.

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